

Symmetry-Adapted Fourier Basis (SAFB) C Library

Documentation and Theory Reference

SAFB Migration Project

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Chapter 1

Introduction to SAFB

1.1 Overview

The Symmetry-Adapted Fourier Basis (SAFB) C library generates scalar fields that strictly adhere to crystallographic space group symmetry constraints. The library is designed for initializing Self-Consistent Field Theory (SCFT) simulations of block copolymer self-assembly, where proper symmetry-adapted initial conditions are critical for convergence.

1.2 Key Concepts

- **Space Group:** A mathematical group of symmetry operations (rotations and translations) that describe the symmetry of a crystal structure. There are 230 space groups in 3D and 17 wallpaper groups in 2D.
- **Star:** A set of symmetry-equivalent reciprocal lattice vectors (Miller indices h,k,l). A star contains all vectors generated by applying the point group operations to a single seed vector.
- **SAFBBasis:** A collection of valid Fourier modes (stars) for a given space group and lattice. Each mode has an associated amplitude and phase determined by symmetry constraints.
- **Field Generation:** The process of computing the real-space field $\psi(\mathbf{r})$ from symmetry-adapted Fourier coefficients using inverse FFT.
- **VTK Export:** Writing the generated field to VTK XML format for visualization in ParaView or similar tools.

1.3 Use Cases

1. **SCFT Initialization:** Generate symmetry-adapted initial conditions for block copolymer self-consistent field theory simulations.

2. **Crystal Structure Analysis:** Study the symmetry properties of different crystal structures (gyroid, BCC, FCC, etc.).
3. **Educational Tool:** Visualize and understand the relationship between space group symmetry and Fourier mode structure.

1.4 Supported Space Groups

The library supports all 230 3D space groups and all 17 2D wallpaper groups. The 3D space groups are stored in `examples/space_groups_3d/` and the 2D wallpaper groups in `examples/space_groups_2d/`.

1.4.1 Notable 3D Space Groups

- **Pm-3m** (No. 221): Simple cubic, used for BCC-like morphologies
- **Ia-3d** (No. 230): Body-centered cubic gyroid, the most studied bicontinuous structure
- **Pn-3m** (No. 224): Diamond structure
- **P4₁32** (No. 213): Chiral cubic structure

1.4.2 Supported 2D Wallpaper Groups

- **Oblique:** P1, P2 (2 groups)
- **Rectangular:** Pm, Pg, Pmm, Pmg, P2gg (5 groups)
- **Square:** P4, P4mm, P4gm, P422 (4 groups)
- **Hexagonal:** P3, P3m1, P31m, P6, P6mm (5 groups)

Chapter 2

Crystallographic Background

2.1 Space Groups

A space group is a mathematical description of the symmetry of a crystal structure. It consists of all symmetry operations (rotations, reflections, inversions, and translations) that leave the crystal invariant.

2.1.1 Symmetry Operations

Each symmetry operation consists of a rotation matrix $\mathbf{R}$ and a fractional translation vector $\mathbf{t}$:

$$\mathbf{r}' = \mathbf{R} \cdot \mathbf{r} + \mathbf{t} \quad (2.1)$$

where $\mathbf{r}$ is a position vector in direct space.

For 3D space groups, $\mathbf{R}$ is a 3×3 integer matrix and $\mathbf{t}$ is a 3-vector of fractions. For 2D wallpaper groups, $\mathbf{R}$ is a 2×2 integer matrix and $\mathbf{t}$ is a 2-vector of fractions (promoted to 3D with z-invariance).

The multiplication of two operations follows:

$$C = AB : \quad \mathbf{R}_C = \mathbf{R}_A \mathbf{R}_B, \quad \mathbf{t}_C = \mathbf{t}_A + \mathbf{R}_A \mathbf{t}_B \quad (2.2)$$

The inverse of an operation $(\mathbf{R}, \mathbf{t})$ is:

$$(\mathbf{R}, \mathbf{t})^{-1} = (\mathbf{R}^{-1}, -\mathbf{R}^{-1} \mathbf{t}) \quad (2.3)$$

2.1.2 Point Group vs. Space Group

- The **point group** consists only of the rotation/reflection matrices (no translations). It describes the rotational symmetry.
- The **space group** includes both rotations and translations. It describes the full symmetry.
- The order of the point group (number of rotation matrices) is always a divisor of the order of the space group (number of symmetry operations).

2.2 Bravais Lattices

2.2.1 The Five 2D Bravais Lattices

There are exactly five distinct Bravais lattice types in two dimensions, classified by their metric properties (lattice parameters a , b , γ):

1. **Oblique:** $a \neq b$, $\gamma \neq 90^\circ$. No symmetry constraints. The most general 2D lattice.
2. **Rectangular Primitive:** $a \neq b$, $\gamma = 90^\circ$. Orthogonal basis with unequal lengths.
3. **Rectangular Centered:** $a \neq b$, $\gamma = 90^\circ$. Additional lattice point at $(a/2, b/2)$. Can be described by a primitive cell with basis vectors $(a/2, -b/2)$ and $(a/2, b/2)$.
4. **Square:** $a = b$, $\gamma = 90^\circ$. Equal orthogonal basis vectors.
5. **Hexagonal:** $a = b$, $\gamma = 120^\circ$. Equal basis vectors at 120° .

2.2.2 Direct Lattice Basis Vectors

The direct lattice is defined by basis vectors $\mathbf{a}_1, \mathbf{a}_2$ (2D) or $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ (3D). Any lattice point is:

$$\mathbf{R} = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 \quad (2D) \quad (2.4)$$

where n_1, n_2 are integers.

The unit cell area in 2D is:

$$A = |\mathbf{a}_1 \times \mathbf{a}_2| = \sqrt{\det(\mathbf{G})} \quad (2.5)$$

2.2.3 Metric Matrix

The metric matrix $\mathbf{G}$ encodes the geometry of the direct lattice:

$$G_{ij} = \mathbf{a}_i \cdot \mathbf{a}_j \quad (2.6)$$

In 2D:

$$\mathbf{G} = \begin{pmatrix} \mathbf{a}_1 \cdot \mathbf{a}_1 & \mathbf{a}_1 \cdot \mathbf{a}_2 \\ \mathbf{a}_2 \cdot \mathbf{a}_1 & \mathbf{a}_2 \cdot \mathbf{a}_2 \end{pmatrix} = \begin{pmatrix} a^2 & ab \cos \gamma \\ ab \cos \gamma & b^2 \end{pmatrix} \quad (2.7)$$

The unit cell area is $A = \sqrt{\det(\mathbf{G})} = ab \sin \gamma$.

2.2.4 Transformation of the Metric Matrix

Under a change of basis $\mathbf{a}'_i = \sum_j \mathbf{a}_j P_{ji}$, the metric matrix transforms as:

$$\mathbf{G}' = \mathbf{P}^T \mathbf{G} \mathbf{P} \quad (2.8)$$

where $\mathbf{P}$ is the transformation matrix. The ratio of cell areas is $A'/A = |\det(\mathbf{P})|$.

2.3 Reciprocal Lattice

2.3.1 Reciprocal Basis Vectors

The reciprocal lattice is the Fourier transform of the direct lattice. For a direct lattice with basis vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$, the reciprocal lattice basis vectors are:

$$\mathbf{b}_1 = 2\pi \frac{\mathbf{a}_2 \times \mathbf{a}_3}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)}, \quad \mathbf{b}_2 = 2\pi \frac{\mathbf{a}_3 \times \mathbf{a}_1}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)}, \quad \mathbf{b}_3 = 2\pi \frac{\mathbf{a}_1 \times \mathbf{a}_2}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)} \quad (2.9)$$

The defining property is:

$$\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi \delta_{ij} \quad (2.10)$$

2.3.2 Reciprocal Lattice Vectors for Each 2D Bravais Lattice

Lattice Type	Direct Basis	Reciprocal Basis
Oblique	$\mathbf{a}_1 = a(1, 0), \mathbf{a}_2 = a(\cos \gamma, \sin \gamma)$	$\mathbf{b}_1 = \frac{2\pi}{a \sin \gamma}(1, -\cot \gamma), \mathbf{b}_2 = \frac{2\pi}{a \sin \gamma}(0, 1)$
Rectangular	$\mathbf{a}_1 = a(1, 0), \mathbf{a}_2 = b(0, 1)$	$\mathbf{b}_1 = \frac{2\pi}{a}(1, 0), \mathbf{b}_2 = \frac{2\pi}{b}(0, 1)$
Centered Rect.	$\mathbf{a}_1 = (a/2, -b/2), \mathbf{a}_2 = (a/2, b/2)$	$\mathbf{b}_1 = \frac{2\pi}{a}(1, 1), \mathbf{b}_2 = \frac{2\pi}{a}(1, -1)$
Square	$\mathbf{a}_1 = a(1, 0), \mathbf{a}_2 = a(0, 1)$	$\mathbf{b}_1 = \frac{2\pi}{a}(1, 0), \mathbf{b}_2 = \frac{2\pi}{a}(0, 1)$
Hexagonal	$\mathbf{a}_1 = a(1, 0), \mathbf{a}_2 = a(1/2, \sqrt{3}/2)$	$\mathbf{b}_1 = \frac{2\pi}{a}(1, -1/\sqrt{3}), \mathbf{b}_2 = \frac{2\pi}{a}(0, 2/\sqrt{3})$

Table 2.1: Direct and reciprocal basis vectors for the five 2D Bravais lattices.

2.3.3 Reciprocal Lattice Metric

The reciprocal metric tensor is:

$$G_{ij}^* = \mathbf{b}_i \cdot \mathbf{b}_j \quad (2.11)$$

It is related to the direct metric by $\mathbf{G}^* = (\mathbf{G}^{-1})^T$ (or equivalently $\mathbf{G}^* = \mathbf{G}^{-1}$ since $\mathbf{G}$ is symmetric).

2.3.4 d-Spacing and Miller Indices

A reciprocal lattice vector is labeled by Miller indices (h, k) :

$$\mathbf{q} = h\mathbf{b}_1 + k\mathbf{b}_2 \quad (2.12)$$

The d-spacing (interplanar distance) is:

$$d_{hk} = \frac{2\pi}{|\mathbf{q}|} = \frac{2\pi}{\sqrt{\mathbf{q}^T \mathbf{G}^* \mathbf{q}}} \quad (2.13)$$

2.4 The 17 Wallpaper Groups

2.4.1 Overview

There are exactly 17 wallpaper groups the distinct symmetry groups of 2D periodic patterns. They are classified into five crystal systems based on their underlying Bravais lattice:

Crystal System	Bravais Lattice	Wallpaper Groups
Oblique	Oblique	p1, p2
Rectangular Primitive	Rectangular Primitive	pm, pg, p2mm, p2mg, p2gg
Rectangular Centered	Rectangular Centered	cm, c2mm
Square	Square	p4, p4mm, p4gm, p422
Hexagonal	Hexagonal	p3, p3m1, p31m, p6, p6mm

Table 2.2: Classification of the 17 wallpaper groups by crystal system.

2.4.2 Symmetry Operations by Group

Group	Operations	Order
p1	Identity only	1
p2	Identity, 2-fold rotation	2
pm	Identity, mirror reflection	2
pg	Identity, glide reflection	2
cm	Identity, centered mirror	2
p2mm	Identity, 2-fold, 2 mirrors, inversion	4
p2mg	Identity, 2-fold, mirror, glide	4
p2gg	Identity, 2-fold, 2 glides	4
c2mm	Identity, 2-fold, 2 mirrors (centered)	4
p4	Identity, 4-fold, 2-fold rotations	4
p4mm	Identity, 4-fold, mirrors	8
p4gm	Identity, 4-fold, mirrors, glides	8
p422	Identity, 4-fold, 2-folds (all directions)	8
p3	Identity, 3-fold rotations	3
p3m1	Identity, 3-fold, mirrors through vertices	6
p31m	Identity, 3-fold, mirrors between vertices	6
p6	Identity, 6-fold rotations	6
p6mm	Identity, 6-fold, mirrors	12

Table 2.3: Symmetry operations for each wallpaper group.

2.4.3 Crystallographic Restriction Theorem

Only n -fold rotations with $n = 1, 2, 3, 4, 6$ are compatible with translational periodicity in 2D. This is the **crystallographic restriction theorem**.

2.4.4 Asymmetric Unit

The **asymmetric unit** is the smallest region of the unit cell that, under all symmetry operations of the group, generates the entire pattern. The area of the asymmetric unit is the unit cell area divided by the order of the group.

2.5 Stars

A **star** of a reciprocal lattice vector $\mathbf{q}$ is the set of all vectors generated by applying the point group operations:

$$\text{Star}(\mathbf{q}) = \{\mathbf{R} \cdot \mathbf{q} \mid \mathbf{R} \in \text{Point Group}\} \cup \{-\mathbf{R} \cdot \mathbf{q} \mid \mathbf{R} \in \text{Point Group}\} \quad (2.14)$$

The union with negatives ensures the star is closed under inversion, which is required for real-valued Fourier series.

2.5.1 Star Properties

- **Multiplicity:** The number of vectors in the star. Determined by the order of the point group divided by the order of the stabilizer subgroup.
- **Closure:** A star is closed if it contains all symmetry-equivalent vectors. The function `star_is_closed()` verifies this.
- **q-squared:** All vectors in a star have the same $|\mathbf{q}|^2$ (q-squared), since rotations preserve length.

2.6 Miller Indices

Miller indices (h, k, l) label reciprocal lattice vectors and crystallographic planes. In the SAFB library, Miller indices are used to:

- Identify and group reciprocal lattice vectors into stars
- Compute the canonical key for lexicographic sorting
- Label modes in the output

The canonical key is computed by sorting the Miller indices and taking the absolute value of the first non-zero element, ensuring a unique identifier for each family of planes.

Chapter 3

Mathematical Formulation

3.1 Symmetry-Adapted Fourier Series

The scalar field $\psi(\mathbf{r})$ is expanded as a Fourier series over symmetry-adapted modes:

$$\psi(\mathbf{r}) = \sum_n A_n \exp(i\mathbf{q}_n \cdot \mathbf{r}) \quad (3.1)$$

where:

- $\mathbf{q}_n$ are the reciprocal lattice vectors (from stars)
- A_n are complex amplitudes constrained by symmetry

The Fourier coefficient for a reciprocal vector $\mathbf{q}$ is:

$$c_{\mathbf{q}} = \frac{1}{\Omega} \int_{\Omega} \psi(\mathbf{r}) \exp(-i\mathbf{q} \cdot \mathbf{r}) d\mathbf{r} \quad (3.2)$$

where Ω is the unit cell area (2D) or volume (3D).

3.2 Phase Constraints from Symmetry

3.2.1 General Phase Relationship

Symmetry imposes constraints on the Fourier coefficients. For a symmetry operation $(\mathbf{R}, \mathbf{t})$ and a reciprocal vector $\mathbf{q}$, the coefficients satisfy:

$$c_{\mathbf{R}\mathbf{q}} = \exp(2\pi i \mathbf{t} \cdot \mathbf{q}) c_{\mathbf{q}} \quad (3.3)$$

This is Theorem B.6 of the PSCF++ manual. It means that once the coefficient for one vector in a star is known, all other coefficients in the star are determined up to a global phase factor.

3.2.2 Phase Factor Computation

The phase factor for a symmetry operation $(\mathbf{R}, \mathbf{t})$ acting on vector $\mathbf{q} = (h, k, l)$ is:

$$\phi = 2\pi \mathbf{t} \cdot \mathbf{q} = 2\pi \sum_j t_j q_j \quad (3.4)$$

where the dot product is computed using fractional coordinates. The exponential $\exp(i\phi)$ gives the phase relationship between symmetry-equivalent coefficients.

3.3 Symmetry-Adapted Fourier Series for Wallpaper Groups

Following Verberck (2012), we derive the symmetry-adapted Fourier series for each wallpaper group. The key insight is that symmetry operations relate Fourier coefficients, reducing the number of independent coefficients.

3.3.1 Rotation Constraints

2-fold Rotation (p2)

For a 2-fold rotation about the origin, $(h, k) \rightarrow (-h, -k)$:

$$c_{h,k} = c_{-h,-k} \quad (3.5)$$

The minimal domain D2 consists of pairs $\{(h, -k), (-h, k)\}$ with $h > 0$, or $h = 0$ and $k > 0$.

3-fold Rotation (p3, hexagonal)

For a 3-fold rotation in the hexagonal lattice:

$$c_{k_1, k_2} = c_{-k_1+k_2, -k_1} = c_{-k_2, k_1-k_2} \quad (3.6)$$

This forms a 3-cycle. The minimal domain D3 reduces coefficients by 3x.

4-fold Rotation (p4, square)

For a 4-fold rotation:

$$c_{k_1, k_2} = c_{-k_2, k_1} = c_{-k_1, -k_2} = c_{k_2, -k_1} \quad (3.7)$$

This forms a 4-cycle. The minimal domain D4 consists of $k_1 > 0$, $0 \leq k_2 < k_1$, reducing coefficients by 4x.

6-fold Rotation (p6, hexagonal)

For a 6-fold rotation:

$$c_{k_1, k_2} = c_{k_2, -k_1+k_2} = c_{-k_1+k_2, -k_1} = c_{-k_1, -k_2} = c_{-k_2, k_1-k_2} = c_{k_1-k_2, k_1} \quad (3.8)$$

This forms a 6-cycle. The minimal domain D6 consists of $0 < k_1$, $0 \leq k_2 < k_1$, reducing coefficients by 6x.

3.3.2 Reflection Constraints

Mirror about x-axis (pm, rectangular)

$$c_{k_1, k_2} = c_{k_1, -k_2} \quad (3.9)$$

The minimal domain is $k_2 \geq 0$, reducing coefficients by 2x.

Mirror about diagonal (cm, centered rectangular)

$$c_{k_1, k_2} = c_{k_2, k_1} \quad (3.10)$$

Mirror about y-axis (p4mm, square)

$$c_{k_1, k_2} = c_{-k_1, k_2} \quad (3.11)$$

3.3.3 Glide Reflection Constraints

Glide reflections introduce additional phase factors of $(-1)^n$ due to the fractional translation component.

Glide along x-axis (pg, rectangular)

$$c_{k_1, k_2} = (-1)^{k_1} c_{k_1, -k_2} \quad (3.12)$$

This implies $c_{k_1, 0} = 0$ for k_1 odd. The phase factor is $h(k_1, k_2) = (-1)^{k_1}$.

Glide along y-axis (p2mg, rectangular)

$$c_{k_1, k_2} = (-1)^{k_2} c_{k_1, -k_2} \quad (3.13)$$

Glide along diagonal (p2gg, rectangular)

$$c_{k_1, k_2} = (-1)^{k_1 + k_2} c_{k_1, -k_2} \quad (3.14)$$

3.3.4 Centering Constraints (cm, c2mm)

For centered rectangular lattices, the centering transformation $\mathbf{r}' = \mathbf{r} + (a/2, b/2)$ leads to:

$$(-1)^{q_1 + q_2} c_{q_1, q_2} = c_{q_1, q_2} \quad (3.15)$$

where $q_1 = k_1 + k_2$, $q_2 = k_1 - k_2$. This implies:

$$c_{q_1, q_2} = 0 \quad \text{for } q_1 + q_2 \text{ odd} \quad (3.16)$$

This is the **centering condition** for the cm and c2mm groups.

For c2mm:

$$c_{q_1, q_2} = c_{-q_1, -q_2} = c_{q_1, -q_2} = c_{-q_1, q_2} = (-1)^{q_1 + q_2} c_{q_1, q_2} \quad (3.17)$$

3.4 Explicit Fourier Expansions

3.4.1 p6 (Hexagonal, 6-fold symmetry)

$$\psi(\mathbf{r}) = c_{0,0} + \sum_{(k_1, k_2) \in D_6} c_{k_1, k_2} \sum_{j=0}^5 \exp(i\mathbf{q}_j \cdot \mathbf{r}) \quad (3.18)$$

where the six vectors $\mathbf{q}_j$ are generated by 6-fold rotation from (k_1, k_2) . Domain D6: $0 < k_1, 0 \leq k_2 < k_1$.

3.4.2 p4 (Square, 4-fold symmetry)

$$\psi(\mathbf{r}) = c_{0,0} + \sum_{(k_1, k_2) \in D_4} c_{k_1, k_2} \sum_{j=0}^3 \exp(i\mathbf{q}_j \cdot \mathbf{r}) \quad (3.19)$$

where the four vectors are (k_1, k_2) , $(-k_2, k_1)$, $(-k_1, -k_2)$, $(k_2, -k_1)$. Domain D4: $k_1 > 0, 0 \leq k_2 < k_1$.

3.4.3 p2 (Oblique/Rectangular, 2-fold symmetry)

$$\psi(\mathbf{r}) = c_{0,0} + \sum_{(k_1, k_2) \in D_2} c_{k_1, k_2} [\exp(i\mathbf{q}_{k_1, k_2} \cdot \mathbf{r}) + \exp(i\mathbf{q}_{-k_1, -k_2} \cdot \mathbf{r})] \quad (3.20)$$

Domain D2: $k_1 > 0$, or $k_1 = 0$ and $k_2 > 0$.

3.4.4 pm (Rectangular, mirror symmetry)

$$\psi(\mathbf{r}) = c_{0,0} + \sum_{k_1} c_{k_1,0} \exp(ik_1 \mathbf{b}_1 \cdot \mathbf{r}) + 2 \sum_{k_1, k_2 > 0} \text{Re} [c_{k_1, k_2} \exp(i(k_1 \mathbf{b}_1 + k_2 \mathbf{b}_2) \cdot \mathbf{r})] \quad (3.21)$$

Domain: $k_2 \geq 0$.

3.4.5 pg (Rectangular, glide symmetry)

$$\psi(x, y) = c_{0,0} + \sum_{k_1 \text{ even}} c_{k_1,0} [\cos(2\pi k_1 x/a) + i \sin(2\pi k_1 x/a)] + 2 \sum_{k_2 > 0} \left\{ \sum_{k_1 \text{ even}} c_{k_1, k_2} [\cos \cos + i \sin \cos] + \sum_{k_1 \text{ odd}} c_{k_1, k_2} [\cos \sin + i \sin \sin] \right\} \quad (3.22)$$

Vanishing condition: $c_{k_1,0} = 0$ for k_1 odd.

3.5 Coefficient Solver Algorithm

The phase constraints form a system of linear equations. For a star with m vectors and r relationships, we have:

$$c_j = e^{i\phi_j} c_k \quad (3.23)$$

for each relationship (j, k, ϕ_j) . The system is solved using BFS (Breadth-First Search) on the relationship graph:

1. Start with a reference coefficient $c_{\text{ref}} = 1.0 + 0i$
2. For each relationship, propagate the phase constraint through the graph
3. Check for consistency (no contradictory phase constraints)
4. All coefficients are determined up to a global phase factor

3.6 Reality Criteria

3.6.1 Centrosymmetric Groups

For wallpaper groups that contain inversion symmetry (2-fold rotation at the origin), all Fourier coefficients $c_{\mathbf{q}}$ must be **real**:

$$c_{\mathbf{q}} \in \mathcal{R} \quad \text{for centrosymmetric groups} \quad (3.24)$$

Centrosymmetric wallpaper groups: p1, p2, pm, p2mm, p2mg, p2gg, c2mm, p4, p4mm, p4gm, p6, p6mm.

3.6.2 Non-Centrosymmetric Groups

For groups without inversion symmetry, the reality condition $c_{-\mathbf{q}} = c_{\mathbf{q}}^*$ relates coefficients within the minimal domain:

$$c_{-\mathbf{q}} = c_{\mathbf{q}}^* \quad (3.25)$$

Non-centrosymmetric wallpaper groups: pg, cm, p3, p3m1, p31m.

3.7 Normalization

The field is normalized using the hyperbolic tangent function to ensure values lie in $[0, 1]$:

$$\psi_{\text{norm}}(\mathbf{r}) = \frac{1}{2} \left[1 + \tanh \left(\frac{\psi(\mathbf{r}) - \langle \psi \rangle}{\sigma \cdot \text{std}(\psi)} \right) \right] \quad (3.26)$$

where σ is a scaling parameter (typically $\sigma = 1.0$).

3.8 Square Norm

The square norm (average of ψ^2 over the unit cell) is:

$$\langle |\psi|^2 \rangle = \frac{1}{V} \int_V |\psi(\mathbf{r})|^2 d\mathbf{r} \quad (3.27)$$

This can be computed numerically on the FFT grid or analytically using Parseval's theorem:

$$\langle |\psi|^2 \rangle = \sum_n |A_n|^2 \quad (3.28)$$

3.9 Fourier Coefficient Maps

Each wallpaper group has a characteristic “map” of equivalent coefficients in (k_1, k_2) space. These maps show:

- **1**: Independent coefficients (one per minimal domain)
- **100**: Coefficients equivalent under symmetry (one value per equivalence class)
- **200**: Doubly-equivalent coefficients (special positions with higher multiplicity)

These maps are complementary to the International Tables for Crystallography and can be used for identifying wallpaper groups from experimental diffraction data.

Chapter 4

Numerical Methods

4.1 FFT-Based Field Generation

The field is computed using the inverse Fast Fourier Transform (FFT):

$$\psi(\mathbf{r}_j) = \mathcal{F}^{-1}\{c_{\mathbf{q}_n}\} \quad (4.1)$$

4.1.1 FFTW Integration

The library uses FFTW (Fastest Fourier Transform in the West) for efficient FFT computation:

- `fftw_plan_dft_3d`: Creates a 3D FFT plan for the specified grid size
- `fftw_execute`: Executes the FFT plan
- `fftw_destroy_plan`: Cleans up the plan

The forward and inverse transforms are:

$$\tilde{f}(\mathbf{q}) = \sum_{\mathbf{r}} f(\mathbf{r}) \exp(-i\mathbf{q} \cdot \mathbf{r}), \quad f(\mathbf{r}) = \frac{1}{N} \sum_{\mathbf{q}} \tilde{f}(\mathbf{q}) \exp(i\mathbf{q} \cdot \mathbf{r}) \quad (4.2)$$

4.1.2 Grid Quadrature

The real-space grid is defined by the lattice parameters and the number of grid points:

$$x_i = \frac{i}{N_x} L_x, \quad y_j = \frac{j}{N_y} L_y, \quad z_k = \frac{k}{N_z} L_z \quad (4.3)$$

where N_x, N_y, N_z are the grid dimensions and L_x, L_y, L_z are the lattice lengths.

For 2D fields, the grid is:

$$x_i = \frac{i}{N_x} L_x, \quad y_j = \frac{j}{N_y} L_y \quad (4.4)$$

The grid spacing determines the maximum resolvable wavevector:

$$q_{\max} = \frac{\pi}{\Delta x} = \frac{N_x \pi}{L_x} \quad (4.5)$$

4.1.3 Reduced Coordinates

The grid points are defined in reduced (fractional) coordinates:

$$r_\alpha = \frac{m_\alpha}{M_\alpha} \quad (4.6)$$

where m_α is the grid index and M_α is the grid size in each direction.

4.2 Coefficient Solver Algorithm

The coefficient solver uses BFS on the star's relationship graph. The algorithm proceeds as follows:

1. Initialize the reference coefficient $c_{\text{ref}} = 1.0 + 0i$
2. Create a queue with the reference index
3. While the queue is not empty:
 - Pop the current index u from the queue
 - For each neighbor (v, ϕ) of u :
 - If v has been visited, verify the phase constraint
 - If v has not been visited, set $c_v = e^{i\phi} c_u$ and add v to the queue

4.2.1 Consistency Checking

After solving, the algorithm verifies that all phase constraints are satisfied:

$$|c_j - e^{i\phi_j} c_k| < \epsilon \quad (4.7)$$

where $\epsilon = 10^{-10}$ is the numerical tolerance. If any constraint is violated, the star is flagged as inconsistent (a “cancelled star” where all coefficients must be zero).

4.2.2 Cancelled Stars

A star is “cancelled” when the symmetry constraints are so restrictive that all coefficients must be zero. This occurs when:

- A centered lattice operation $(I, \mathbf{t})$ with $\mathbf{t} \neq 0$ produces systematic cancellations
- The phase factor $e^{i\phi} = -1$ for a self-referential constraint, forcing $c = -c = 0$

4.3 Memory Layout

The library uses flat arrays for efficiency:

- Reciprocal space: flat array of complex coefficients, indexed by star and vector position
- Real space: flat array of real field values, indexed by (i, j, k) in row-major order
- The mapping between reciprocal and real space is handled by FFTW

4.4 Numerical Precision

Quantity	Tolerance
Metric matrix round-trip	10^{-10}
Phase factors	10^{-8}
Field values (relative)	10^{-5}
Coefficient solver	10^{-10}

Table 4.1: Numerical tolerances used in SAFB validation.

4.5 FFT Aliasing

The discrete Fourier transform introduces aliasing effects. Components with wavevectors exceeding the Nyquist frequency are folded back into the grid. The SAFB library handles this by:

- Only including stars with $|\mathbf{q}|$ within the grid's resolution
- Using grid sizes that are multiples of the symmetry order (e.g., N divisible by 4 for square symmetry)
- Verifying that all star vectors map to valid grid points

Chapter 5

Library Architecture

5.1 Module Overview

Module	Purpose	Key Functions
domain.h/c	Data structures	LatticeInfo, Star, SAFBBasis
symmetry_ops.h/c	Symmetry operations	read_spacegroup_ops_txt, star_from_hkl
basis.h/c	Basis construction	family_planes_info, build_basis
initializers.h/c	Amplitude assignment	random_initialization, manual_initialization
field.h/c	Field generation	build_field, build_field_2d
engine.h/c	High-level API	engine_create, engine_init_random
analytic.h/c	Analytical calculations	calculate_square_norm

Table 5.1: Library modules and their responsibilities

5.2 Data Structures

5.2.1 LatticeInfo

A structure holding lattice parameters and metric tensors:

- Lx, Ly, Lz: Lattice lengths (Å)
- alpha, beta, gamma: Lattice angles (degrees)
- G[3][3]: Reciprocal metric tensor
- Ginv[3][3]: Inverse metric tensor
- dim: Dimension (2 or 3)

5.2.2 Star

A structure holding star information:

- `hkl[3]`: Canonical Miller indices
- `q2:  $|\mathbf{q}|^2$` (q-squared)
- `multiplicity`: Number of vectors in the star
- `closed`: 1 if closed under inversion
- `vectors[3][MAX_STAR_SIZE]`: Symmetry-equivalent vectors
- `vector_keys[MAX_STAR_SIZE]`: Lexicographic keys
- `phase_factors[MAX_STAR_SIZE]`: Phase factors

5.2.3 SAFBBasis

A structure holding the complete basis:

- `n_modes`: Number of modes (families)
- `modes[MAX_MODES]`: Mode data
- `centrosymmetric`: 1 if inversion is in point group
- `inversion_at_origin`: 1 if inversion at origin

5.3 API Usage

5.3.1 Basic Workflow

1. Create engine from space group file and lattice parameters
2. Initialize (random or manual)
3. Build field (inverse FFT)
4. Output field to VTK
5. Clean up

5.3.2 Example Code

```
// Create engine from space group file
Engine *engine = engine_create(
    "examples/space_groups_3d/ia_3d.txt",
    "examples/lattice_cubic.txt");

// Random initialization with seed 42
engine_init_random(engine, 42);

// Build field on 64x64x64 grid
engine_build_field(engine, 64, 64, 64);

// Output field to VTK
engine_output_field(engine, "gyroid.vts");

// Clean up
engine_destroy(engine);
```

Chapter 6

User Manual

6.1 Building the Library

```
source /sandbox/setup_build.sh    # Activate build environment
make                               # Build all test programs
make test                         # Run all tests
```

6.2 Running Examples

6.2.1 3D Example: Ia-3d Gyroid

```
# Build the library first
source /sandbox/setup_build.sh
make

# Run the gyroid example (uses space group file)
./examples/run_gyroid.sh
```

6.2.2 2D Example: P4 Square Lattice

```
# Run the 2D P4 example
./examples/run_2d_p4.sh
```

6.3 Configuration

6.3.1 Lattice Parameters

Lattice parameters are specified in `examples/lattice_cubic.txt`:

```
a=10, b=10, c=10, alpha=90, beta=90, gamma=90
```

For non-cubic lattices, modify the values accordingly.

6.3.2 Grid Resolution

The grid resolution is specified in the engine call:

```
engine_build_field(engine, Nx, Ny, Nz)
```

Typical values: $64 \times 64 \times 64$ for production, $16 \times 16 \times 16$ for testing.

6.4 Output Files

- `*.vts`: VTK XML structured point data files (viewable in ParaView)
- `*.dat`: Field data in plain text format
- `*.json`: Basis metadata (mode count, q-values, etc.)

Chapter 7

Examples and Workflows

7.1 Gyroid (Ia-3d)

The gyroid is a bicontinuous minimal surface with Ia-3d symmetry. It is one of the most studied structures in block copolymer science.

7.1.1 Generation Steps

1. Load space group operations from `examples/space_groups_3d/ia_3d.txt`
2. Load lattice parameters (cubic, $a = 10$)
3. Build basis: find all valid stars up to $N_{\max} = 5$
4. Random initialization: assign random amplitudes with symmetry constraints
5. Build field: compute inverse FFT on 64^3 grid
6. Normalize: apply tanh normalization
7. Export: write to `gyroid.vts`

7.1.2 Expected Results

- 5 modes (families) with multiplicities: 8, 6, 24, 12, 48
- Total Fourier modes: 98
- Field range: $[0, 1]$ after normalization
- Square norm: ≈ 0.5 for random initialization

7.1.3 Symmetry Analysis

The Ia-3d space group has 16 symmetry operations. The gyroid structure is characterized by:

- No reflection planes (chiral structure)
- Three-fold screw axes along $\langle 111 \rangle$ directions
- Two-fold rotation axes along $\langle 110 \rangle$ directions
- Inversion center at the origin

The gyroid minimal surface is defined by:

$$\cos(x + y) \cos(y + z) + \cos(y + z) \cos(x + z) + \cos(x + z) \cos(x + y) = 0 \quad (7.1)$$

7.2 BCC-like (Pm-3m)

The Pm-3m space group generates BCC-like density modulations.

7.2.1 Generation Steps

Same as gyroid, but with `pm3m.txt` space group file and cubic lattice.

7.2.2 Expected Results

- Simple cubic symmetry with mirror planes in all three directions
- Field shows spherical density maxima at BCC lattice points
- Useful for studying sphere-forming block copolymers

7.3 2D Examples

7.3.1 Square Lattice (P4)

The P4 wallpaper group generates 4-fold symmetric patterns.

- 4 symmetry operations (0° , 90° , 180° , 270° rotations)
- Supports 2D field generation
- Useful for studying 2D self-assembly patterns

```
# Run the 2D P4 example
source /sandbox/setup_build.sh
make
./examples/run_2d_p4.sh
```

7.3.2 Hexagonal Lattice (P6)

The P6 wallpaper group generates 6-fold symmetric patterns.

- 6 symmetry operations (0° , 60° , 120° , 180° , 240° , 300°)
- Supports 2D field generation
- Hexagonal packing patterns

```
# Run the 2D P6 example
source /sandbox/setup_build.sh
make
./examples/run_2d_p6.sh
```

7.3.3 Rectangular Lattice (Pmm2)

The Pmm2 wallpaper group generates patterns with two perpendicular mirror lines.

- 4 symmetry operations (identity, two mirrors, 2-fold rotation)
- Rectangular Bravais lattice
- Useful for studying anisotropic patterns

7.4 Workflow Summary

The typical workflow is:

1. **Setup:** Source the build environment and compile the library
2. **Configure:** Set lattice parameters and grid resolution
3. **Initialize:** Choose an initialization method (random, manual, or file-based)
4. **Generate:** Compute the real-space field via inverse FFT
5. **Visualize:** Export to VTK format and view in ParaView

7.5 Parameter Tuning

7.5.1 Grid Resolution

Higher grid resolution gives more accurate fields but increases computation time:

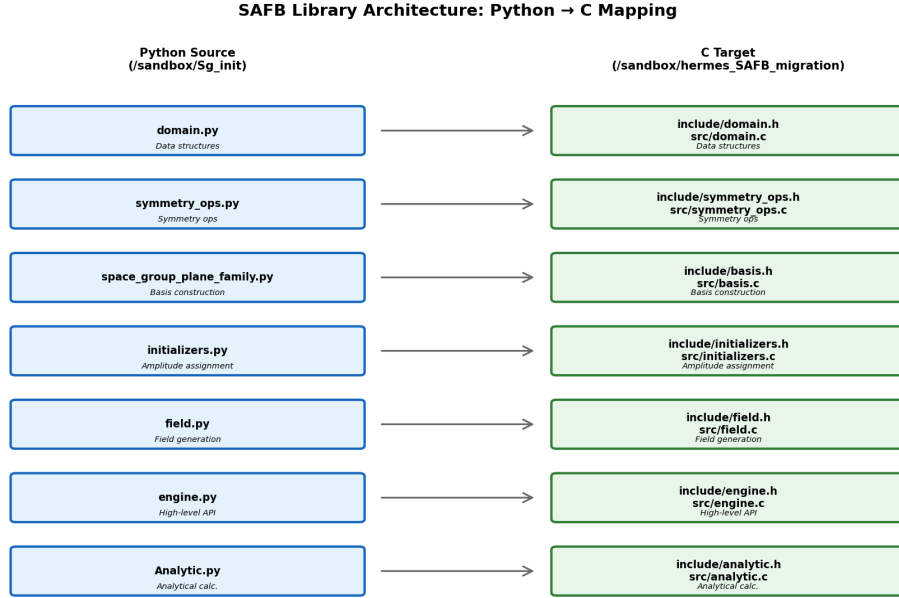


Figure 7.1: SAFB library architecture: Python-to-C module mapping.

Resolution	3D Time (est.)	Use Case
16^3	$\sim 0.1\text{s}$	Testing, development
32^3	$\sim 0.5\text{s}$	Quick prototyping
64^3	$\sim 2\text{s}$	Production runs
128^3	$\sim 15\text{s}$	High-precision studies

Table 7.1: Estimated computation times for different grid resolutions.

7.5.2 Number of Stars

More stars (higher N_{max}) give more accurate representations but increase the number of free parameters:

- $N_{\text{max}} = 3$: Few modes, smooth fields, fast convergence
- $N_{\text{max}} = 5$: Moderate resolution, typical for SCFT
- $N_{\text{max}} = 8$: High resolution, for detailed studies

7.5.3 Scaling Parameter

The tanh normalization parameter σ controls the sharpness of the field:

- $\sigma = 0.5$: Smooth transitions, low contrast
- $\sigma = 1.0$: Moderate contrast (default)
- $\sigma = 2.0$: Sharp interfaces, high contrast

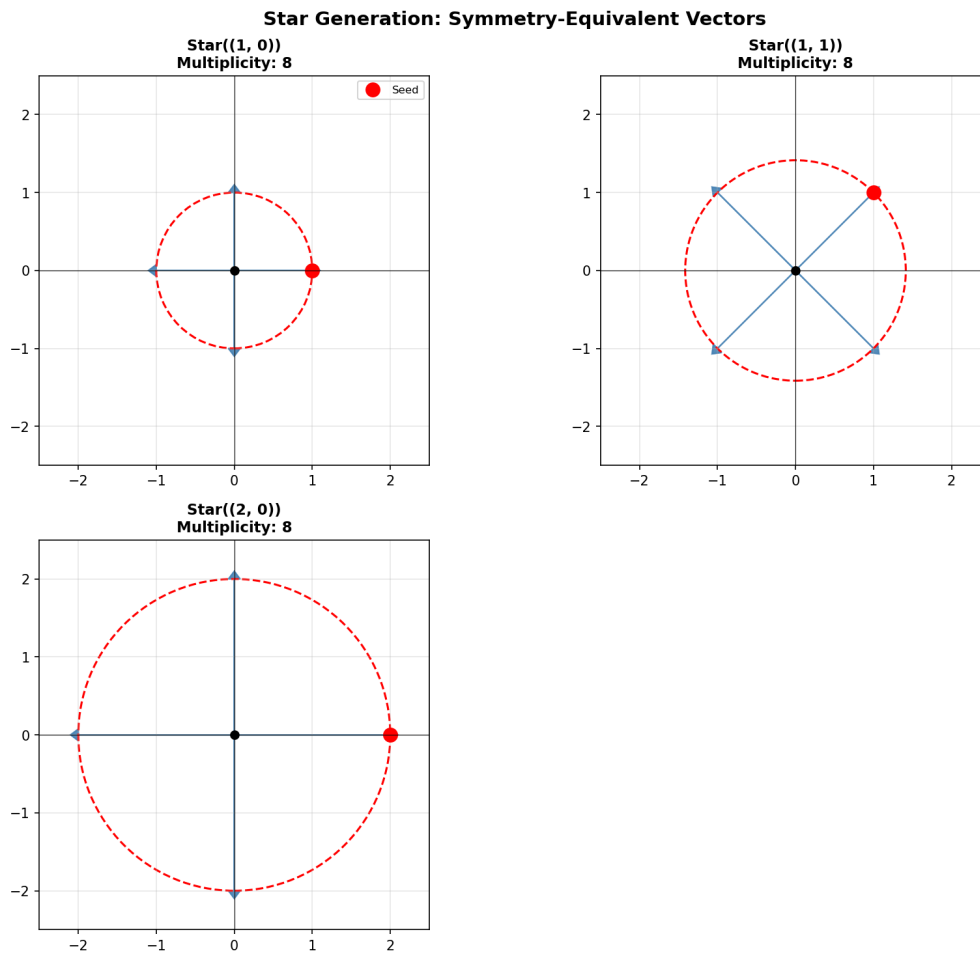


Figure 7.2: Star generation: symmetry-equivalent vectors in reciprocal space.

Chapter 8

Glossary

Glossary of Crystallographic Terms

Space Group A mathematical group describing all symmetry operations (rotations, reflections, inversions, and translations) that leave a crystal structure invariant. There are 230 space groups in 3D and 17 wallpaper groups in 2D.

Wallpaper Group A 2D space group describing the symmetry of patterns that repeat in two dimensions. Also called plane groups. There are 17 wallpaper groups.

Point Group The set of rotational and reflectional symmetry operations of a crystal, ignoring translations. The point group describes the rotational symmetry of the crystal class.

Symmetry Operation A transformation (rotation, reflection, inversion, or translation) that maps the crystal onto itself. Each operation is represented by a rotation matrix $\mathbf{R}$ and a fractional translation vector $\mathbf{t}$.

Miller Indices Integer triplet (h, k, l) that labels a reciprocal lattice vector or a family of crystallographic planes. The indices are the reciprocals of the intercepts of the plane with the crystallographic axes.

Star A set of symmetry-equivalent reciprocal lattice vectors. A star contains all vectors generated by applying the point group operations to a seed vector, plus their negatives.

Multiplicity The number of vectors in a star. Determined by the order of the point group divided by the order of the stabilizer subgroup of the seed vector.

Reciprocal Lattice The Fourier transform of the direct (real-space) lattice. Reciprocal lattice vectors $\mathbf{q} = h\mathbf{b}_1 + k\mathbf{b}_2 + l\mathbf{b}_3$ label the Fourier modes in the crystal.

Direct Lattice The real-space lattice defined by basis vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$. The crystal structure is a periodic arrangement of atoms on this lattice.

Metric Tensor A symmetric matrix $\mathbf{G}$ encoding the geometry of the reciprocal lattice: $G_{ij} = \mathbf{b}_i \cdot \mathbf{b}_j$. Used to compute $|\mathbf{q}|^2$ for any Miller index triplet.

Phase Constraint A relationship between Fourier coefficients of symmetry-equivalent vectors: $c_{\mathbf{R}\mathbf{q}} = \exp(2\pi i \mathbf{t} \cdot \mathbf{q}) c_{\mathbf{q}}$. Ensures the field respects space group symmetry.

Fourier Mode A single term in the Fourier expansion of the field: $A_n \exp(i\mathbf{q}_n \cdot \mathbf{r})$. Each mode corresponds to a reciprocal lattice vector.

SAFBBasis A collection of valid Fourier modes (stars) for a given space group and lattice. Contains all the information needed to generate a symmetry-adapted field.

Field Generation The process of computing the real-space field $\psi(\mathbf{r})$ from Fourier coefficients using inverse FFT.

Normalization The process of scaling the field to a standard range, typically $[0, 1]$, using a tanh function.

Square Norm The average of ψ^2 over the unit cell: $\langle |\psi|^2 \rangle = \frac{1}{V} \int_V |\psi|^2 d\mathbf{r}$. Measures the overall amplitude of the field.

Centrosymmetric A crystal structure is centrosymmetric if it contains an inversion center. In terms of the point group, this means the inversion matrix $(-I)$ is a member.

Inversion at Origin A special case where the inversion operation maps $\mathbf{r} \rightarrow -\mathbf{r}$ (no translation). This simplifies the phase constraints.

Canonical Key A lexicographic identifier for a family of planes, computed by sorting the Miller indices and taking the absolute value of the first non-zero element.

Closed Star A star that contains all symmetry-equivalent vectors, including all negatives. Required for real-valued Fourier series.

BFS Coefficient Solver A breadth-first search algorithm that propagates phase constraints through the star's relationship graph to determine all Fourier coefficients.

FFTW Fastest Fourier Transform in the West. A widely-used library for computing FFTs. The SAFB library uses FFTW for efficient field generation.

VTK Visualization Toolkit. A file format for scientific visualization data. The SAFB library exports fields in VTK XML structured point data format.

SCFT Self-Consistent Field Theory. A mean-field theory for polymer self-assembly. The SAFB library provides symmetry-adapted initial conditions for SCFT simulations.

Block Copolymer A polymer consisting of two or more different polymer blocks covalently bonded together. Block copolymers self-assemble into periodic nanostructures.

Gyroid A bicontinuous minimal surface with Ia-3d symmetry. One of the most studied structures in block copolymer science.

BCC Body-centered cubic. A crystal structure with space group Pm-3m. Block copolymers can form BCC-like density modulations.

FCC Face-centered cubic. A crystal structure with space group Fm-3m.

Diamond A crystal structure with space group Pn-3m. Block copolymers can form diamond-like structures.

Wallpaper Group P1 The simplest 2D space group, containing only the identity operation. No symmetry beyond translation.

Wallpaper Group P6 A 2D space group with 6-fold rotational symmetry. Generates hexagonal patterns.

Wallpaper Group P4 A 2D space group with 4-fold rotational symmetry. Generates square patterns.

Chapter 9

References